System Test Cases

One test case per supported command

October 2026

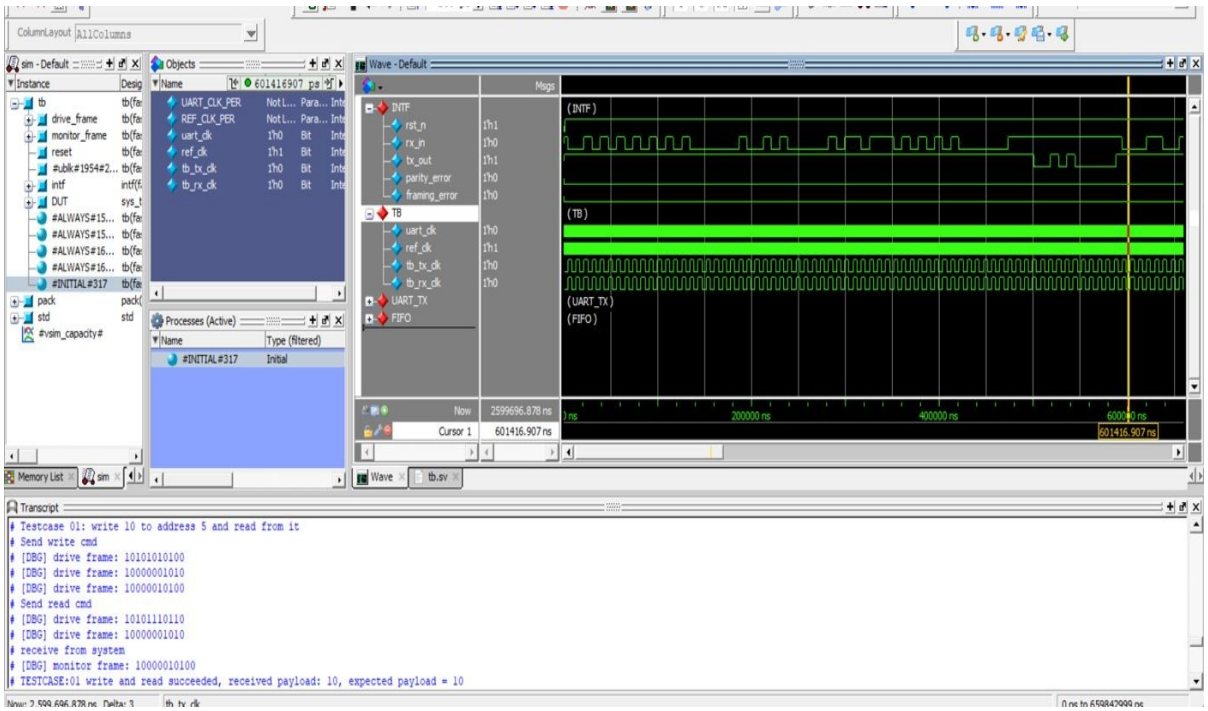
Test Case Summary

Four test cases, one for each supported command, run against SYS_TOP in Questa Sim. Frame format shown MSB first: stop bit, parity bit, 8 data bits, start bit. Parity is enabled, even (default configuration).

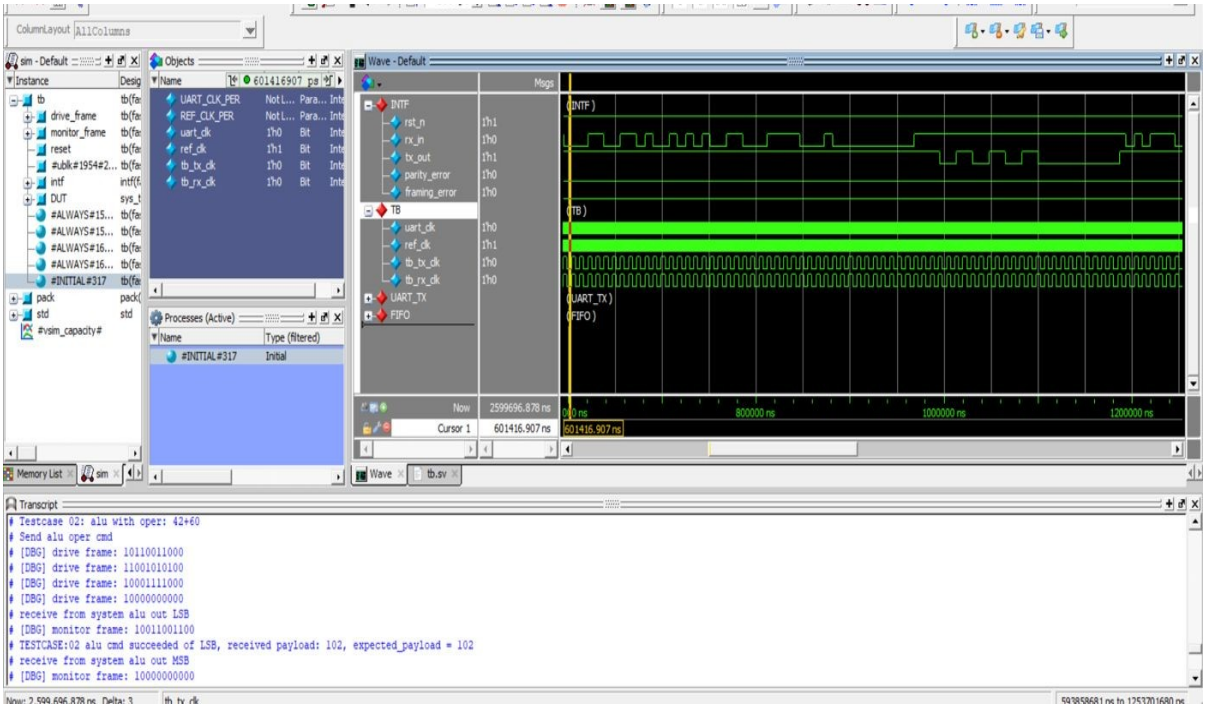
ID	Command	Stimulus	Frames driven on RX_IN (stop,par,data,start)	Expected	Observed	Result
TC1	RF Write (0xAA)	Frames: 0xAA, addr 0x05, data 0x0A (10)	10101010100 10000001010 10000010100	Register 5 holds 10 (checked by TC2 read-back)	Write accepted; value returned by TC2	PASS
TC2	RF Read (0xBB)	Frames: 0xBB, addr 0x05	10101110110 10000001010	1 frame returned on TX_OUT with payload 10	Frame 10000010100 received, payload = 10	PASS
TC3	ALU with operands (0xCC)	Frames: 0xCC, A=42, B=60, FUN=0 (ADD)	10110011000 11001010100 10001111000 10000000000	42+60 = 102 -> LSB 102, MSB 0	LSB = 102, MSB = 0	PASS
TC4	ALU no operands (0xDD)	Frames: 0xDD, FUN=2 (MUL) using A=42,B=60 already in REG0/REG1	10110111010 11000000100	42*60 = 2520 (0x09D8) -> LSB 216, MSB 9	LSB = 216, MSB = 9	PASS

Test cases TC1 and TC2 come from one simulation (write then read of address 5); TC4 uses the operands written by TC3.

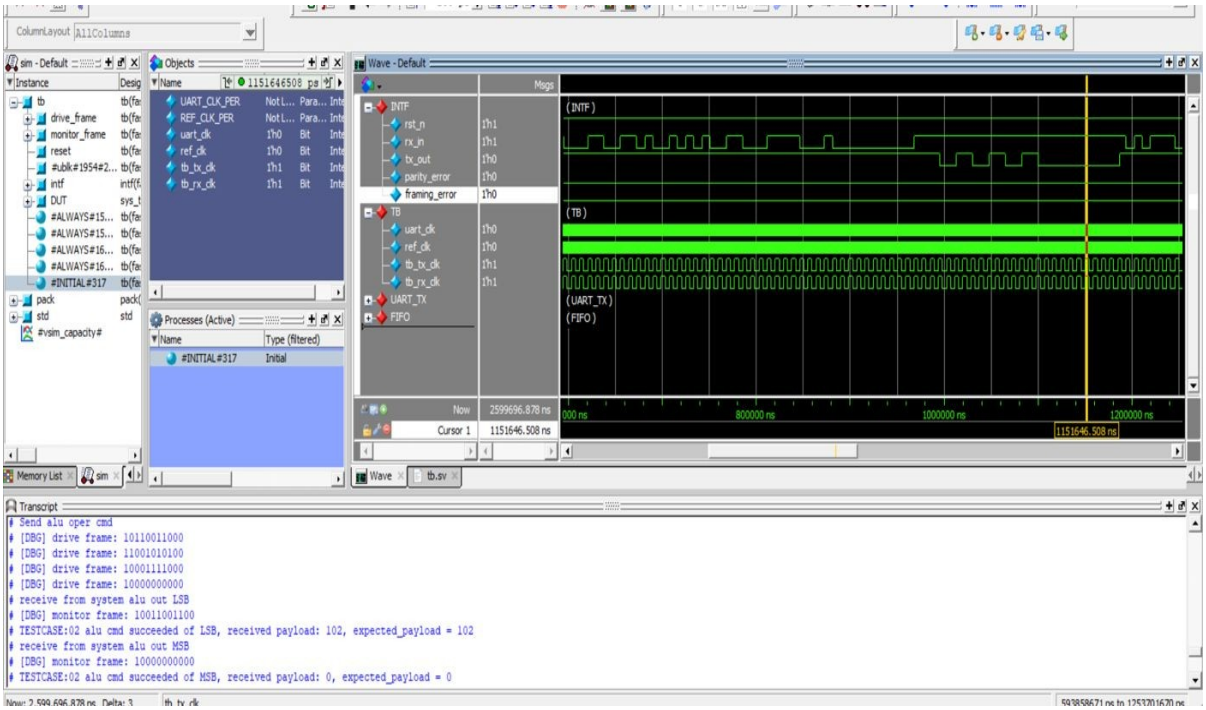
Test Case Evidence



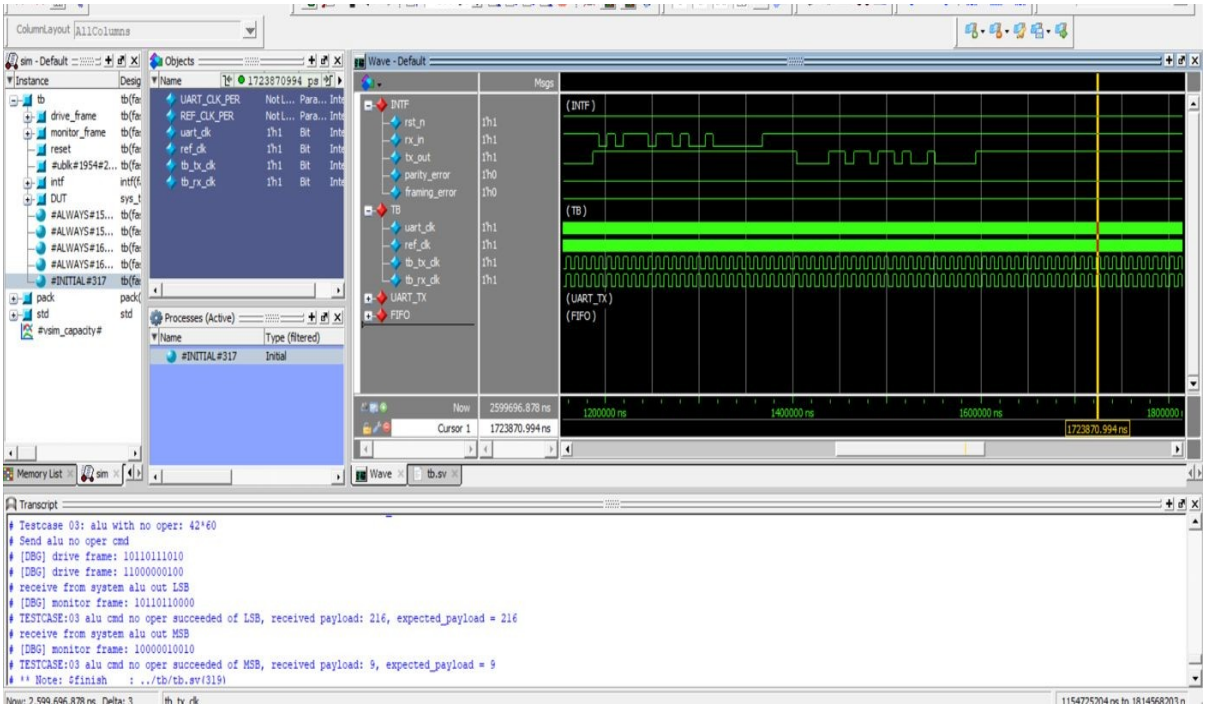
TC1 + TC2 - RegFile write 10 to address 5, then read (payload 10 = expected 10)



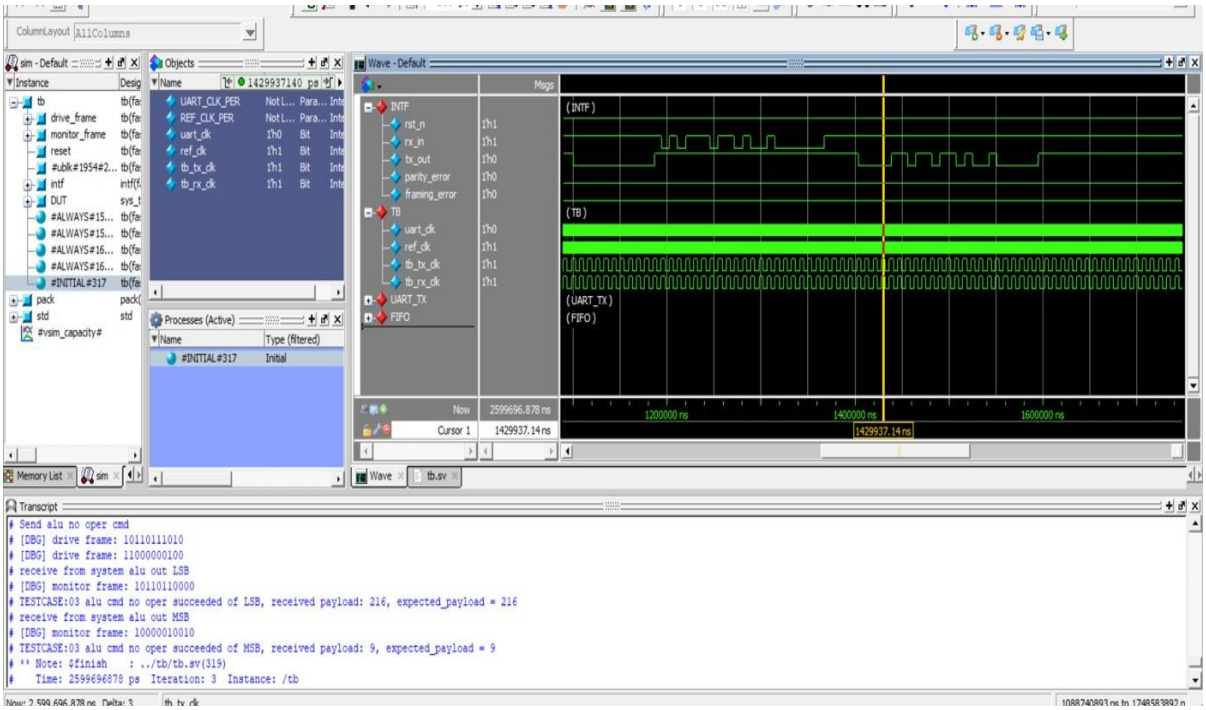
TC3 - ALU with operands 42+60



TC3 - result LSB 102, MSB 0 (pass)



TC4 - ALU without operands (FUN=2, multiply)



TC4 - result LSB 216, MSB 9 (pass), $finish